PHYS 2325.001 – Fall 2025 – Course Syllabus

1. Course Information

Course Number / Section: PHYS 2325-001
Course Title: Mechanics
Term: Fall 2025
Days and Times: Tues/Thurs 11:30 am – 12:45 pm
Location: SCI 1.220

2. Instructor Contact Information

Instructor: Dr. Zihao Ou
E-mail address: Zihao.Ou@UTDallas.edu
Office location: SCI 2.186
Office hours and location: 3 pm – 4 pm on Wednesdays

TA Office Hours (name, email, time, location):

Email Tas to schedule office hours upon requests.

Primary TA:

Rohit Kajla (rohit.kajla@utdallas.edu)

Sahithi Nadella (sahithinadella@utdallas.edu)

Homework assistance and tutoring will be provided by a pool of TAs, as specified above. You can attend any TA's office hours, whether they are the primary TA for our section or not.

3. Course Description

3 semester credit hours. Calculus based. Basic physics includes the study of space and time, kinematics, forces, energy and momentum, conservation laws, rotational motion, torques, harmonic oscillation, and waves. Two lectures per week.

4. Student Learning Objectives / Outcomes

1. Uncertainties, significant figures and scientific notations; addition and subtraction of vectors; scalar and vector product of vectors.

2. Concepts of time, position, velocity and acceleration in 1D; mathematical relations relating them.

3. Velocity and acceleration in 2D and 3D; circular motion and relative velocity.

4. Concept of force; three Newton's Laws relating force with motion.

5. Application of Newton's Laws in particle equilibrium and dynamics.

6. Concepts of work and energy, and work-energy theorem.

7. Concepts of momentum; momentum conservation in collision problems.

8. Velocity, acceleration, and energy in rotational motion.

9. Relating force with motion in rotational motion and with equilibrium.

10. Newton's law of gravitation; relating gravitational force with motion.

11. Simple harmonic motion.

12. Mathematical description of waves, including velocity, period, frequency, phase, energy and superposition.

5. Textbook Recommendation

Textbook

Modified Mastering Physics for University Physics with Modern Physics, 15th Edition Author(s): Young, Hugh | Freedman, Roger, Textbook ISBN-13: 9780135159705. The 11th-14th editions will work just fine. I will teach and prepare the course materials with the assumption that you don't have access to the textbook and the lecture slides will be available through e-learning after each lecture.

6. Tentative Schedule

Dates	Lecture	Chapter	HW	HW Due
8/26	Physical quantities and vectors	1	HW1	9/2
8/28	Physical quantities and vectors	1		
9/2	1D kinematics	2	HW2	9/9
9/4	1D kinematics	2		
9/9	2D kinematics	3	HW3	9/16
9/11	2D kinematics	3		
9/16	Newton's laws of motion	4	HW4	9/23
9/18	Newton's laws of motion	4		
9/23	Review of Chapter 1-4	1-4	No HW	-
9/25	Exam #1 (11:30 AM - 12:30 PM) SCI 1.220			
9/30	Application of Newton's Laws	5	HW5	10/7
10/2	Application of Newton's Laws	5		
10/7	Work and Kinetic Energy	6	HW6	10/14
10/9	Potential Energy and Energy Conservation	7		
10/14	Review for Chapter 5-7	5-7	No HW	-
10/16	Exam #2 (11:30 AM - 12:30 PM) SCI 1.220			
10/21	Momentum and Collisions	8	HW7	10/28
10/23	Rotational Motion	9		
10/28	Rotational Dynamics	10	HW8	11/4
10/30	Equilibrium and Elasticity	11		
11/4	Review for Chapter 8-11	8-11	No HW	-
11/6	Exam #3 (11:30 AM - 12:30 PM) SCI 1.220			
11/11	Fluid Mechanics	12	HW9	11/18
11/13	Fluid Mechanics	12		
11/18	Gravitation	13	HW10	12/2
11/20	Periodic Motion	14		
12/2	Mechanical Waves	15	HW11	12/9
12/4	Sound and Hearing	16		
12/9	Review for Chapter 12-16	11-16	-	-
12/11	Exam #4 (5:00 PM - 6:00 PM) ECSW 1.315			

7. Grading Policy

Total grades are based on a combination of the items below.

Homework	25 points
Exams	75 points (25 points for each exam; the final score = sum of the top three grades of the four exams)
Pretest/posttest	4 points (extra credits)
TOTAL	**104 points** (with 4 extra credit points)

Grading Scale

A+ (> 97), A (93 - 96.9), A- (90 – 92.9), B+ (87 – 89.9), B (83 – 86.9), B- (80 - 82.9)

C+ (77 - 79.9), C (73 - 76.9), C- (70 - 72.9), D+ (67 – 69.9), D (63 – 66.9), D- (60- 62.9), F (< 60)

Example

If your grades break down as follows:

20 out of 25 points for the homework.

The four exam scores are 72, 80, 100, and 88 (full score is 100 points), you will get 67 out of 75 points for exams.

4 extra points for taking pretest/posttest.

Then your total score will be 20+4+67 = 91 and your letter grade will be A- (because 91 lies in between 90 and 92.9, the range for letter grade A-).

8. Course & Instructor Policies

1. Explanation on Homework (25 points equally distributed among 12 assignments)

Homework will be given via the e-learning platform with the due time displayed in the homework system. No late homework is accepted under normal circumstances since answers will become available after due date. Attendance is not mandatory, but students will be expected to participate in regular classroom activities.

2. Explanation on 4 Exams (75 points)

The 4 exams will be given, each of 25 points (including the final exam). Exams will be hosted in the classroom. They will be in the closed book. You can see preliminary exam dates from the calendar above. The exam dates are subject to change. If anything changes, announcements will be made in class.

Exams will involve multiple choice and fill-in-blank problems, with same format as homework in the E-learning system.

Formulas sheet will be provided a couple of days before exams. You must know the concepts and vocabulary for the exams. Exams will be mainly based on in-class examples and homework.

Each exam will have 25 points contribution to the final grade. Your grade of the 4 exams will be the sum of the top 3 grades amongst 4. The maximum possible score for the 4 exams is 75. For example, if your grades for the 4 exams are 10, 20, 25, and 22, then you will get 20+25+22=67 (out of 75 points).

If you miss one of the four exams, your grade will be the sum of the other three exams taken. If you miss two of them, your final grade will only include two exams taken. If you miss three of them, your final grade will only have one exam take, which contributes to 25 in maximum.

None of the 4 exams will cover the material before the previous exam. For example, Exam #2 will cover the lectures after exam #1 and before exam #2. The exams are based on lectures and homework.

Make-up exams will only be given under exceptional circumstances with well-documented reasons beyond the students' control. Requests should be made to the instructor at least one week before each scheduled exam. Students that need this arrangement are expected to justify the reason for your absence. Makeup exams may have different problem sets from the original exams.

During exams:

Please, be kind to your classmates (and the instructor) and avoid interruptions by turning off your cell phones, laptops, and other electronics during lectures (those with documented medical needs are exempt from this requirement). Also, you are asked to arrive and leave on time.

Valid picture ID (Comet card) must be presented for exams. These will be checked.

Calculators will be necessary for all exams. Graphing calculators and programmable calculators will not be allowed in the exams. Calculators should not have text functions. A little scientific calculator that has trig functions can be obtained very inexpensively ($10-$20) and should be all that is used during exams.

After exams:

Any question about an exam grade must be addressed within two weeks after the grade is posted. After that all grades are final.

3. Explanation on Pretest/Posttest (4 credit points)

You have the opportunity to do two quizzes as part of your introductory physics course. The quizzes consist of multiple-choice questions and are useful to the department in gathering information about the effectiveness of our courses. There are two extra credit points associated with the two tests. You will receive a credit point by simply taking each quiz—your grade will NOT depend on your performance. There is no penalty for not taking these two quizzes.

Test window for the two quizzes (The exact name for the exam section on eLearning sites might be slightly different depending on the display settings that are selected by eLearning):

Instructor	Course Information	Exam Name	Exam Start Date	Exam End Date	Exam Duration (Min)
Paul MacAlevey	PHYS 1301 / PHYS 2325.All Sections	Mechanics Pretest	1/21/2025	1/31/2025	60
		Mechanics Posttest	4/16/2025	4/26/2025	60

The quizzes will be taken at the testing center on the first floor of the Synergy Park North 2 building (SPN2). Students register for the quizzes at https://ets.utdallas.edu/testing- center. If you really need to do the test at the OSA test center, please send your accommodation letter to Dr. Paul MacAlevey, paulmac@utdallas.edu, and he can help you with this request. The test is delivered through eLearning, but on an eLearning site (with Dr. Paul MacAlevey as the instructor) separate from the one used for lectures of this class. The quizzes are not available to be done without an in-person proctor.

The test center requires students to reserve the test time online **at least 48 hours before the intended exam time**. This means that students cannot register at least two days before the Exam End Date on the table above. For example, pretests close on Friday Jan 31st and the Test Center closes at 5:30 that day. The Test Center wants exams to end at 5:15 so a 1- hour test must begin at or before 4:15 that day. The last time to register with the Test center for such a test is 4:15 two days before. This means that the 'window' for registering for the pretest closes at 4:15 on Wednesday 29th.

Please refer to the testing center guidelines on how to schedule, reschedule, cancel your test and how to take it during your scheduled time: https://ets.utdallas.edu/testing- center/students/. Tests are unavailable when the test center is not open (open hours are listed in the test center page: https://ets.utdallas.edu/testing-center) or when the test center is fully reserved.

These tests help inform the department what you had known before taking the class and how much you have progressed after taking the class. So please avoid guessing at answers. Your grade on these two quizzes will NOT depend on your performance, you will earn the credit by simply taking the quizzes.

Each quiz will finish 1 hour after you click 'Begin Assessment'. You must complete the quiz in a single interval of 1 hour or less.

Class Materials

The instructor may provide class materials that will be made available to all students registered for this class as they are intended to supplement the classroom experience. These materials may be downloaded during the course; however, these materials are for registered students' use only. Classroom materials may not be reproduced or shared with those not in class or uploaded to other online environments except to implement an approved Office of Student Accessibility accommodation. Failure to comply with these University requirements is a violation of the Student Code of Conduct.

Class Attendance

The University's attendance policy requirement is that individual faculty set their course attendance requirements. Regular and punctual class attendance is expected. Students who fail to attend class regularly are inviting scholastic difficulty. In some courses, instructors may have special attendance requirements; these should be made known to students during the first week of classes.

Class Participation

Regular class participation is expected. Students who fail to participate in class regularly are inviting scholastic difficulty. A portion of the grade for this course is directly tied to your participation in this class. It also includes engaging in group or other activities during class that solicit your feedback on homework assignments, readings, or materials covered in the lectures (and/or labs). Class participation is documented by faculty.
Successful participation is defined as consistently adhering to University requirements, as presented in this syllabus. Failure to comply with these University requirements is a violation of the Student Code of Conduct.

Class Recordings

Students are expected to follow appropriate University policies and maintain the security of passwords used to access recorded lectures. Unless the Office of Student Access Ability has approved the student to record the instruction, students are expressly prohibited from recording any part of this course. Recordings may not be published, reproduced, or shared with those not in the class, or uploaded to other online environments except to implement an approved Office of Student Access Ability accommodation. Failure to comply with these University requirements is a violation of the Student Code of Conduct.

Comet Creed

This creed was voted on by the UT Dallas student body in 2014. It is a standard that Comets choose to live by and encourage others to do the same:

"*As a Comet, I pledge honesty, integrity, and service in all that I do*."

Academic Support Resources

The information contained in the following link lists the University's academic support resources for all students.

Please see http://go.utdallas.edu/academic-support-resources.

UT Dallas Syllabus Policies and Procedures

The information contained in the following link constitutes the University's policies and procedures segment of the course syllabus. Please review the catalog sections regarding the credit/no credit or pass/fail grading option and withdrawal from class.

Please go to http://go.utdallas.edu/syllabus-policies for these policies.